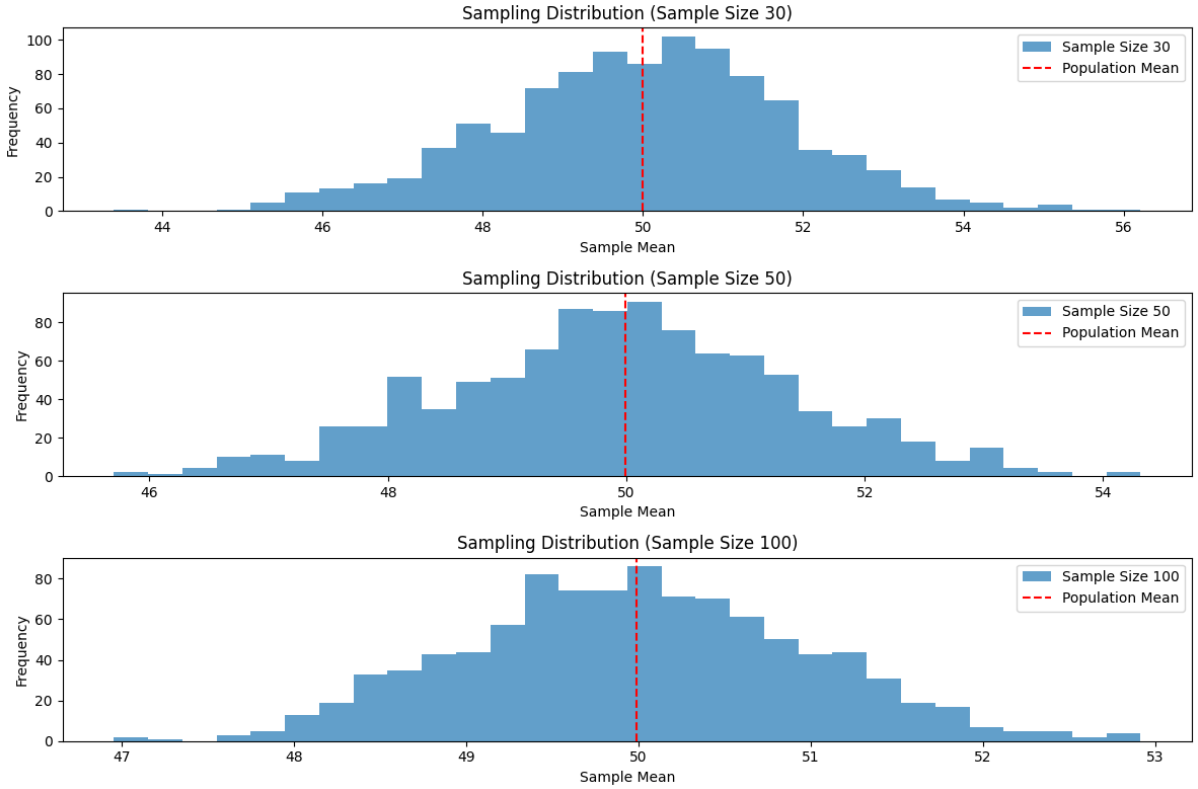
17) Random Sampling and Sampling Distribution.

Program:

```
import numpy as np
import matplotlib.pyplot as plt
population_mean = 50
population_std = 10
population_size = 100000
population = np.random.normal(
    population_mean, population_std, population_size
)
sample_sizes = [30, 50, 100]
num_samples = 1000
sample_means = {}
for size in sample_sizes:
    sample_means[size] = []
    for _ in range(num_samples):
        sample = np.random.choice(population, size=size, replace=False)
        sample_means[size].append(np.mean(sample))
plt.figure(figsize=(12, 8))
for i, size in enumerate(sample_sizes):
    plt.subplot(len(sample_sizes), 1, i + 1)
    plt.hist(sample_means[size], bins=30, alpha=0.7,
             label=f'Sample Size {size}')
    plt.axvline(np.mean(population), color='red',
                linestyle='dashed', linewidth=1.5,
                label='Population Mean')
```

```
        plt.title(f'Sampling Distribution (Sample Size {size})')
        plt.xlabel('Sample Mean')
        plt.ylabel('Frequency')
        plt.legend()
    plt.tight_layout()
    plt.show()
```

Output:



Result:

Thus, random sampling and sampling distributions were successfully generated and visualized for different sample sizes.